Kaicheng Dong

Email: 12012612@mail.sustech.edu.cn
Phone: +49 15226938330

Research Interests

Arithmetic geometry; Arakelov geometry; adelic line bundles and height theory; non-archimedean analytic geometry; Bogomolov-type conjectures; arithmetic intersection theory; algebraic curves and their reductions.

Education

MSc in Mathematics – ALGANT Master Program
Algebra, Geometry and Number Theory
University of Bordeaux, France 2025–2026
University of Regensburg, Germany 2024–2025
Current grade: 1.28, German scale

BSc in Mathematics and Applied Mathematics
Southern University of Science and Technology, China 2020–2024
Honors Program in Mathematics
GPA: 8.56 / 10

Master's Thesis

Admissible Pairing and Effective Bogomolov Results
Supervisor: Professor Pascal Autissier, University of Bordeaux 2026

The thesis studies Zhang's theory of admissible pairings on curves and reformulates part of it in the language of adelic line bundles. It focuses on admissible metrics at archimedean and non-archimedean places, metrized graphs, local Néron symbols, admissible adelic line bundles, and their relation to effective Bogomolov-type results.

Research Experience

Undergraduate Research Project
Serre's Conjecture II and Galois Cohomology
Supervisor: Professor Hu Yong
Southern University of Science and Technology 2023–2024

Relevant Coursework and Training

Diophantine Geometry
University of Regensburg 2024–2025
Topics included heights, equidistribution, Bogomolov-type problems.

Complex Geometry
University of Bordeaux 2025–2026
A year-long course including Hodge theory, the Hard Lefschetz theorem, the weak Lefschetz theorem, Kodaira vanishing, and related topics.

Further background
Algebraic geometry; arithmetic surfaces; intersection theory; Berkovich analytic spaces; Picard varieties and Jacobians.

Talks

Arithmetic Bigness and Uniform Bogomolov-type Theory
ALGANT Alumni in China Conference, University of Science and Technology of China 2025
Slides: `https://faculty.ustc.edu.cn/yqliang/files/ALGANTSlides/25DongKaicheng.pdf`

Seminars and Academic Activities

Organizer
Reading Seminar on Algebraic Fibered Surfaces 2026
Based on Qing Liu, *Algebraic Geometry and Arithmetic Curves*.
The seminar focused on intersection theory on fibered surfaces, regular models, semistable reduction, and the Deligne–Mumford theory of stable reduction of curves.

Participant and Speaker
Reading Seminar on *Formes différentielles réelles et courants sur les espaces de Berkovich*
Organized by Mingchen Xia

- Presented Sections 11.3–11.4.
- Topics included local properties of Berkovich analytic spaces and morphisms.

Conferences and Schools

- ALGANT Alumni in China – "20 Years of ALGANT" 2025
- Summer School on Arithmetic Geometry, Westlake University 2024

Scholarships and Grants

- Erasmus+ Mobility Grant, European Union 2025–2026
- Honors Program in Mathematics, Southern University of Science and Technology

Publications

No publications at this stage.

Languages

- Chinese: native
- English: fluent